ChestX-AI Assistant: Evidence-Grounded Chest X-Ray Reporting with CNNs and Constrained LLMs

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***Abstract*—Chest radiography is the most frequently performed imaging examination worldwide, but expert interpretation is time-consuming and exhibits inter-observer variability. We present ChestX-AI Assistant, a modular pipeline that couples a convolutional neural network (CNN) for multi-label thoracic disease detection with a constrained large language model (LLM) for grounded report generation. The vision module is trained on 133,186 frontal chest radiographs from 45,173 patients drawn from the CheXpert dataset, with a strict patient-level 70/15/15 split to prevent leakage. We compare a ResNet-18 baseline, which obtains a mean test AUROC of 0.7510 after 15/15 epochs, against a DenseNet-121 model, which obtains a mean test AUROC of 0.7527 and early stops at epoch 14/15, using identical data, augmentation, and weighted binary cross-entropy under the U-Zeros uncertainty policy. To bridge raw probabilities and human-readable text without inviting hallucination, we introduce an Evidence JSON contract: a structured artifact containing per-class probabilities, top findings, and image-quality flags that the downstream LLM is contractually limited to reference. Grad-CAM saliency provides per-prediction interpretability. The constrained LLM produces a four-section structured response with calibrated hedging language tied to probability bands. The response is then fused with the Evidence JSON by a ReportLab rendering layer into a single-page clinician-facing structured PDF report that carries the disclaimer, patient information panel, and color-coded probability table alongside the LLM narrative. This preserves the audit trail between probabilities and prose. We further compare three LLM backends, OpenAI, Gemini, and Groq, on the same Evidence JSON. The work demonstrates that a clear data contract between a vision model and a generative model, extended through the rendering layer, is a practical mechanism for safer AI-assisted radiology.**

***Index Terms*—Chest radiography, multi-label classification, deep learning, CheXpert, DenseNet-121, ResNet-18, Grad-CAM, large language models, retrieval-augmented generation, structured PDF reporting, clinical decision support.**

I. Introduction

Chest radiography (CXR) is the most commonly ordered medical imaging study, used for screening, triage, and follow-up of cardiopulmonary disease. Despite its ubiquity, accurate interpretation depends on subspecialty expertise that is not uniformly available, and reading volumes routinely exceed comfortable per-radiologist throughput. Computer-aided diagnosis has therefore emerged as an active area of research, with deep convolutional neural networks (CNNs) demonstrating strong performance on selected thoracic findings [1]–[3].

Two practical gaps limit deployment of these models in clinical workflows. First, raw per-class probabilities are not directly actionable for clinicians and are often unintelligible to patients; a probability vector is not a report. Second, report generation systems can produce fluent but factually incorrect output, which is an unacceptable failure mode in medicine [4], [5]. The maturity of large language models (LLMs), including few-shot language models, makes this issue more important because unconstrained models may hallucinate findings unsupported by visual evidence [6], [7].

We argue that the right architectural primitive is not an end-to-end multimodal model but a strict data contract between a deterministic vision module and a generative language module. Concretely, we propose a six-stage pipeline: (i) a CNN that emits model probabilities for five thoracic findings, (ii) a Grad-CAM module that produces saliency evidence, (iii) an Evidence JSON object that summarizes model outputs with image-quality flags, (iv) a constrained LLM that is prompted to write only what the Evidence JSON supports, (v) an optional retrieval-augmented generation (RAG) layer that supplies approved phrasing templates, and (vi) a ReportLab rendering step that fuses the JSON and the LLM narrative into a single-page clinician-facing structured PDF report.

Our contributions are threefold. First, we benchmark a ResNet-18 baseline and a DenseNet-121 improved model on a strict patient-level split of CheXpert under the U-Zeros uncertainty policy, achieving mean test AUROCs of 0.7510 and 0.7527 respectively across five pathologies. Second, we formalize an Evidence JSON contract that operationally separates visual prediction from language generation and prevents the LLM from emitting findings unsupported by the CNN. Third, we provide an open, reproducible reference implementation spanning data preparation, training, evaluation, Grad-CAM, three interchangeable LLM backends, and a ReportLab-based PDF deliverable. The reference implementation is available at https://github.com/Jaiswal-Akshay/ChestX-AI-Assistant.

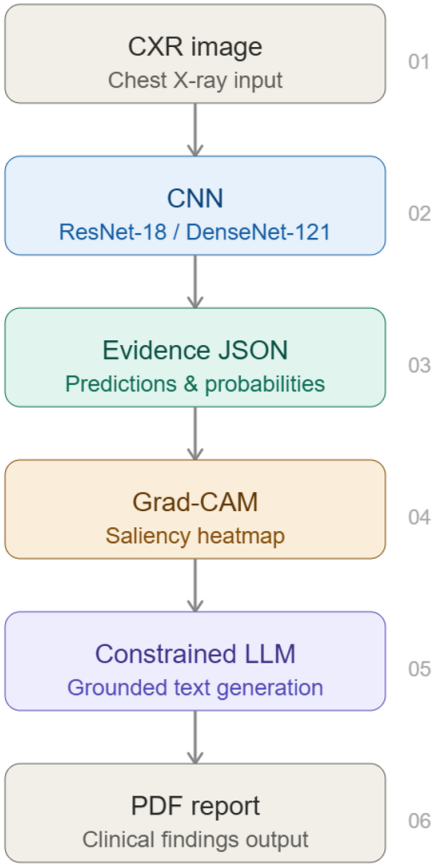


Fig. 1: Overview of the ChestX-AI Assistant pipeline: CXR image → CNN → Evidence JSON → Grad-CAM → constrained LLM → PDF report.

II. Literature Review

This section reviews prior work on chest X-ray analysis, including large-scale datasets, CNN-based disease detection, explainability methods, and recent advances in language-guided report generation.

A. Large-Scale CXR Datasets and Weak Supervision

Progress in CXR modeling has been driven by large public datasets. ChestX-ray14 released over 100,000 frontal radiographs labeled for 14 findings via natural-language processing of radiology reports, establishing a widely used benchmark [8]. CheXpert extended this paradigm to 224,316 radiographs of 65,240 patients and introduced explicit uncertainty labels [2]. MIMIC-CXR subsequently paired more than 370,000 CXRs with free-text reports, enabling cross-modal research [9]. A consistent observation across these benchmarks is that labels are silver labels mined from text rather than manually annotated pixel-level labels, so models must tolerate label noise. CheXpert++ explored differentiable, probabilistic labelers to make weak supervision faster and more reliable [10].

B. CNN and Vision-Language Models for Thoracic Disease Detection

DenseNet, ResNet, and EfficientNet families have all been applied to multi-label CXR classification. CheXNet, a 121-layer DenseNet trained on ChestX-ray14, reported radiologist-level pneumonia detection [1]. DenseNet-121 is therefore a natural reference architecture for CheXpert-style tasks [11]. ResNet architectures remain useful cost-bounded baselines because of their strong transfer-learning performance and simpler structure [12]. More recent work has explored self-supervised and vision-language pretraining for medical imaging, including expert-level pathology detection from unannotated X-rays and biomedical image-text learning [3], [13], [14]. These approaches show the value of language-image alignment, but for safety we keep the classifier-to-language boundary explicit through Evidence JSON.

C. Explainability via Gradient-Based Saliency

Trust in clinical AI requires interpretability. Gradient-weighted class activation mapping, or Grad-CAM, produces class-discriminative saliency maps that highlight regions most influential to a CNN's prediction [15]. While saliency is not proof of causal reasoning, it provides a useful auditing tool: maps that activate outside the lung fields, on tubes, or on text annotations are early warnings of confounded features. We integrate Grad-CAM as part of the evidence package presented to the language model.

D. Radiology Report Generation

End-to-end report generation has progressed from CNN-RNN encoder-decoders to memory-augmented transformers [16] and knowledge-grounded systems that target clinical efficacy rather than text similarity [17]. Other work has explored relational memory and reinforcement learning for more robust radiology report generation [18]. A recurring lesson is that BLEU and ROUGE correlate poorly with diagnostic correctness, motivating evaluation by extracting clinical labels from generated text and comparing them to ground truth. This insight directly informs our design choice to ground the LLM in structured evidence rather than raw image embeddings.

E. LLM Hallucination and Retrieval Grounding

Modern LLMs produce fluent medical prose [7], but hallucination is well documented and dangerous in clinical contexts [4]. Retrieval-augmented generation mitigates hallucination by conditioning generation on retrieved documents [5]. General-purpose representation-learning systems such as CLIP further show the power of language-supervised visual learning, but medical use requires careful grounding and validation [19]. Our Evidence JSON contract serves a similar grounding function: it is the only set of claims the LLM is permitted to make.

F. Identified Gap

Despite these advances, few open implementations cleanly separate the visual evidence layer from the language layer. Most demonstrations either stop at probabilities or deliver fluent but ungrounded text. The gap we target is the operational interface between a CXR classifier and an LLM: what should the classifier hand to the language model, and what is the language model contractually forbidden to invent? Our Evidence JSON proposes a concrete answer.

III. Methods

This section describes the dataset, preprocessing pipeline, model architectures, training procedure, and the proposed CNN–Evidence–LLM framework used for multi-label chest X-ray analysis and grounded report generation.

A. Dataset and Patient-Level Split

We use the CheXpert v1.0 small release [2], restricting to frontal projections, PA and AP, only. After removing studies with no available frontal view and applying quality filtering, we obtain 191,027 images. The five target labels are Atelectasis, Cardiomegaly, Consolidation, Pleural Effusion, and Pneumonia.

Because CheXpert contains multiple studies per patient, a naive random split can leak information across folds. We split at the patient level into 70% train, 15% validation, and 15% test, using a fixed random seed of 42. The resulting split sizes and per-class prevalence are summarized in Table I. Pneumonia prevalence is approximately 2.5% across all splits, while Pleural Effusion appears in approximately 40% of studies, creating an imbalance ratio of roughly 16:1.

TABLE I: Patient-Level Split Statistics on CheXpert Frontal Subset

Split	Images	Patients	At.	Ca.	Co.	PE	Pn.
Train	133,186	45,173	.155	.123	.068	.402	.025
Val	29,244	9,680	.158	.121	.066	.401	.023
Test	28,597	9,681	.157	.122	.071	.406	.025

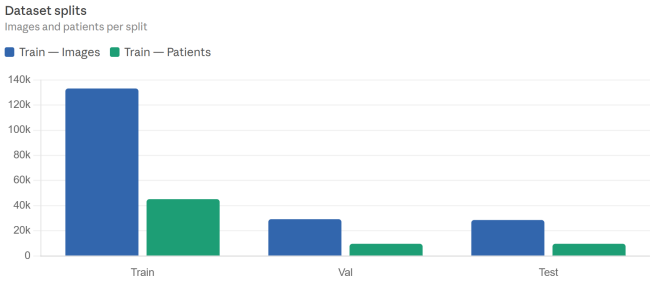


Fig. 2: Dataset split distribution showing the number of images and unique patients in the train, validation, and test sets.

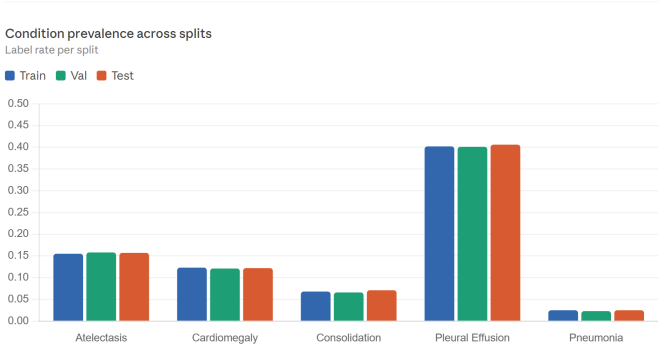


Fig. 3: Condition prevalence across train, validation, and test splits. Pleural Effusion is the most common label, while Pneumonia is the rarest.

B. Uncertainty Policy

CheXpert labels take values in $\{1, 0, -1, blank\}$, where -1 marks reports flagged as uncertain. Two standard policies map this to a binary multi-label target: U-Zeros, where $-1 \to 0$, and U-Ones, where $-1 \to 1$. Because U-Zeros is more conservative and does not invent positive cases from ambiguous text, we adopt U-Zeros as our default. Both NaN and unmapped values are also treated as 0.

C. Image Preprocessing and Augmentation

All images are loaded as single-channel grayscale, resized to 224×224 with aspect ratio preserved by padding, and normalized using ImageNet statistics, $\mu = 0.485$ and $\sigma = 0.229$ [20]. The single channel is replicated to match the pretrained backbone input distribution. Pixel-intensity histograms from preliminary analysis confirm that CheXpert images are pre-windowed but not pre-normalized, justifying this step.

Quality filtering removes images smaller than 100 px, with standard deviation below 5.0, or contrast below 15.0 based on the 95th–5th percentile spread. These thresholds were tuned on a 1,000-image audit and discard fewer than 1% of studies. On the training fold only, we apply random horizontal flip with $p = 0.5$, small random rotations of $\pm 10°$, and mild brightness/contrast jitter.

D. Models

We compare two architectures, both pretrained on ImageNet and fine-tuned end-to-end.

ResNet-18 baseline: The first convolution is replaced with a one-channel input convolution, while the final fully connected layer is replaced with a five-way linear head. ResNet-18 is fast, has approximately 11M parameters, and serves as a cost-bounded baseline. [12]

DenseNet-121 improved model: Following the CheXNet convention, DenseNet-121 is adapted to single-channel input and a five-output classification head. DenseNet's dense connectivity tends to be parameter-efficient on medical imaging tasks. [11]

E. Training Procedure

Both models are optimized with Adam [21] using learning rate 3×10^{-4} and weight decay 1×10^{-4} for up to 15 epochs with batch size 16. We use binary cross-entropy with logits, BCEWithLogitsLoss, with per-class positive weights set to N_{neg}/N_{pos} on the training split. A ReduceLROnPlateau scheduler reduces the learning rate when validation loss stalls. Early stopping with patience 5 is monitored on validation mean AUROC, and the best checkpoint is retained for evaluation.

F. Evaluation Protocol

Our primary metric is per-class area under the ROC curve, AUROC, reported individually and as a macro mean. AUROC is threshold-independent and robust to imbalance, both important for the prevalence gap between Pleural Effusion and Pneumonia. Validation AUROC drives model selection, and the held-out test set is evaluated once per model after hyperparameters are frozen.

G. Grad-CAM Saliency

For each top finding with probability ≥ 0.5, we compute Grad-CAM using the last convolutional block of the chosen backbone. For ResNet-18 this is `layer4`; for DenseNet-121 this is `features.denseblock4`. The resulting low-resolution attribution map is upsampled to 224×224, normalized to $[0, 1]$, and overlaid on the original image. We also compute a coarse spatial summary, such as "saliency concentrated in left lower zone," and append this textual summary to the Evidence JSON.

H. Evidence JSON Contract

The bridge between vision and language is a structured JSON object containing per-class probabilities, top findings, image-quality flags, and optional Grad-CAM region summaries. The downstream LLM prompt forbids the model from referencing disease names not present in the JSON, inventing numerical confidences, or upgrading hedging language from possible to definite.

```
{
  "findings": {
    "Atelectasis": 0.4973,
    "Cardiomegaly": 0.7973,
    "Consolidation": 0.2037,
    "Pleural Effusion": 0.0358,
    "Pneumonia": 0.3853
  },
  "top_findings": [
    ["Cardiomegaly", 0.7973],
    ["Atelectasis", 0.4973]
  ],
  "image_quality": "adequate"
}
```

Listing 1: Example Evidence JSON

I. Constrained LLM Report Generation

The Evidence JSON is fed to an LLM via a system prompt enforcing four rules: (1) reference only diseases listed in the JSON; (2) translate probabilities to hedging language using the discrete map $p \geq 0.70$ as "likely," $0.40 \leq p < 0.70$ as "possible," and $p < 0.40$ as "no strong support"; (3) emit four sections: Findings, Impression, Limitations, and Patient Summary; and (4) always include a non-diagnostic disclaimer. We implement and compare a deterministic Python rule-based generator and three LLM backends: OpenAI, Google Gemini, and Groq. The rule-based generator is the canonical fallback when no API key is configured.

J. PDF Deliverable Rendering

The LLM's structured response, together with the original Evidence JSON, is rendered into a clinician-facing structured PDF report using ReportLab. The PDF is the final shareable artifact of the pipeline and embeds six elements on a single letter-sized page: a branded title block with report date; a patient information panel with name, MRN, date of birth, exam type, and image-quality flag; the submitted chest X-ray thumbnail; a color-coded probability table; the LLM-generated Findings and Impression; and a red disclaimer band with an auto-generated footer. The Evidence JSON drives every quantitative cell on the page, while the LLM text is confined to prose blocks.

K. Reproducibility

All configuration, including seed, learning rate, weight decay, scheduler, label list, uncertainty policy, image size, and augmentation list, is captured in YAML files committed to the repository. At training time the resolved configuration is serialized alongside the model checkpoint as `config_used.yaml`, and per-epoch metrics are appended to `metrics.csv`. The README documents environment setup, dataset acquisition through Stanford's portal, and the four-command pipeline: prepare, train, evaluate, and report.

IV. Results

A. Quantitative Performance

Table II reports per-class AUROC, mean AUROC, test loss, and training-completion status for both models on the held-out 28,597-image test set. DenseNet-121 attains a mean AUROC of 0.7527, a 0.18-point improvement over ResNet-18, while reducing test BCE loss from 1.0058 to 0.9928. Both models perform best on Pleural Effusion and Cardiomegaly. Atelectasis is the weakest class for both models, consistent with its subtle and overlapping visual presentation. The largest per-class gain is on Pneumonia, suggesting DenseNet's deeper feature reuse helps most under extreme imbalance.

TABLE II: Per-Class Test AUROC, Test Loss, and Training Status

Metric	ResNet-18	DenseNet-121	Δ
Epochs run	15/15	14/15 ES	–
Atelectasis	0.6464	0.6423	-0.004
Cardiomegaly	0.8547	0.8537	-0.001
Consolidation	0.7118	0.7169	+0.005
Pleural Effusion	0.8562	0.8558	0.000
Pneumonia	0.6858	0.6949	+0.009
Mean AUROC	0.7510	0.7527	+0.002
Test BCE loss	1.0058	0.9928	-0.013
Training status	Completed	ES @ ep. 14	–

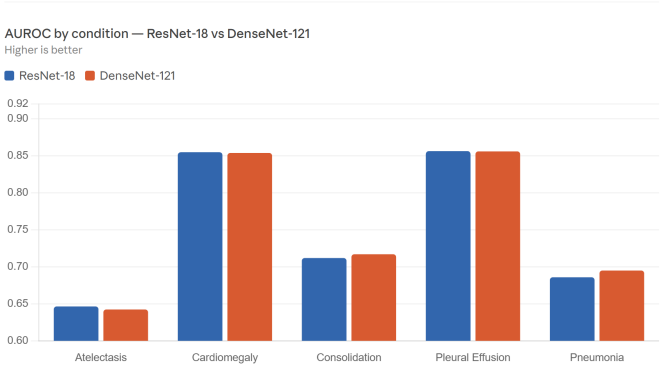


Fig. 4: Per-class AUROC comparison between ResNet-18 and DenseNet-121. DenseNet-121 improves most on Consolidation and Pneumonia, while both models perform strongest on Cardiomegaly and Pleural Effusion.

B. Training Dynamics and Early-Stopping Behavior

ResNet-18 ran to the full 15-epoch budget, whereas DenseNet-121 triggered the patience-5 early-stopping criterion on validation mean AUROC after epoch 14. Its best checkpoint was recorded at epoch 9 with validation mean AUROC 0.7513. ResNet-18 peaked later at epoch 12 with validation mean AUROC 0.7505. DenseNet therefore reaches peak generalization earlier and achieves a slightly better mean test AUROC with lower test BCE loss. Training was performed primarily on CPU with a partial GPU run, which constrained the experiment matrix.

C. Comparison with Prior Work

Direct comparison to CheXNet is complicated by differences in dataset, label set, and evaluation protocol. CheXNet was evaluated on ChestX-ray14 with 14 labels, whereas this project evaluates five CheXpert labels on a large patient-level frontal-only test set. The original CheXpert paper reports per-class AUROCs on a smaller radiologist-labeled validation set and uses both frontal and lateral views [2]. Within our patient-level split and U-Zeros protocol, DenseNet-121's mean AUROC of 0.7527 is a reasonable reproduction under compute and scope constraints.

D. Qualitative Output: Generated Report

Listing 2 reproduces a rule-based report generated from the Evidence JSON in Listing 1. The output respects all four prompt rules: only trained diseases appear, hedging language matches probability bands, the four-section structure is preserved, and the Limitations section disclaims diagnostic intent.

```
FINDINGS
Atelectasis is possibly present (p=0.50).
Cardiomegaly is likely present (p=0.80).
No strong support for consolidation (0.20).
No strong support for effusion (0.04).
No strong support for pneumonia (0.39).

IMPRESSION
Likely findings: Cardiomegaly.
Additional possible findings: Atelectasis.

LIMITATIONS
This is an AI decision-support output
generated from CNN probabilities only.
It is not a diagnosis and should not be
used as a standalone clinical decision tool.
Clinical correlation is recommended.

PATIENT SUMMARY
The AI system thinks the strongest finding
is cardiomegaly. This is only a support
result and should be reviewed by a clinician.
```

Listing 2: Generated Structured Report

E. Cross-Backend LLM Consistency

Replacing the rule-based generator with OpenAI, Gemini, or Groq backends produced reports with the same factual content when given identical Evidence JSON and identical system prompts. The reports preserved the same diseases and hedging tier per disease, while differing stylistically. No backend introduced a finding absent from the Evidence JSON in a manual review of 20 generated reports. This is anecdotal rather than a controlled hallucination study, but it supports the design hypothesis that the contract, not the backend, is the primary safeguard.

This comparison is qualitative and based on a limited manual review, and should not be interpreted as a formal hallucination benchmark.

F. PDF Deliverable: Why a Document, Not a Probability Vector

The end-to-end pipeline terminates not at a probability vector and not at a text blob, but at a single-page PDF that a clinician can drop into a chart and a patient can take home. This format change is consequential. First, the PDF restores the audit chain: the same probabilities emitted by the CNN are visible in a color-coded table on the same page as the prose that interprets them. Second, the disclaimer travels with the document wherever it is forwarded or printed. Third, the PDF surfaces evidence flags, patient metadata, and exam context that are absent from a raw probability list. We view JSON-to-PDF rendering not as cosmetic but as a continuation of the same anti-hallucination contract.

V. Discussion

A. What the Numbers Mean Clinically

An AUROC of approximately 0.86 for Pleural Effusion means that, given a random pair of images, one with effusion and one without, the model assigns a higher probability to the positive image about 86% of the time. This may be useful for triage prioritization, but it is below the threshold for autonomous reporting. The AUROC of approximately 0.64 on Atelectasis would require calibrated thresholding and radiologist review. The Limitations section of every generated report communicates this caveat.

B. Why the Evidence JSON Matters

The contract decouples three failure modes that are usually entangled: classifier error, translation error, and hallucination. Classifier error becomes measurable through AUROC. Translation error becomes unit-testable because the rule-based generator is deterministic. Hallucination risk is narrowed because the LLM has no access to the image and is instructed to ignore unlisted diseases. We do not claim the contract eliminates hallucination, but it shifts hallucination from inventing diseases to misusing hedging language, which is a narrower failure class.

C. Limitations

Several constraints qualify these results. First, most training was performed on CPU, and a stronger GPU setup could enable longer training, larger batches, and more extensive hyperparameter search. Second, we evaluate five labels rather than the full CheXpert 14. Third, all training and evaluation use CheXpert; external validation on ChestX-ray14 or MIMIC-CXR would test domain transfer. Fourth, no clinical user study was performed. Fifth, we report AUROC but not Brier score or reliability diagrams, so the probability-to-language map assumes calibration that has not been fully validated.

D. Threats to Validity

Label noise from the CheXpert labeler is the largest threat. Preliminary EDA showed all-zero labels on images with visible pathology, suggesting that reported AUROCs may be lower bounds on performance under perfect labeling. Confounding from medical hardware such as pacemakers, ICDs, and tubes is a second threat. Grad-CAM is intended to surface such behavior, but we have not yet performed a systematic audit of saliency-map distribution outside the lung fields.

VI. Conclusion and Future Work

We presented ChestX-AI Assistant, a modular pipeline that pairs a DenseNet-121 multi-label CXR classifier with a constrained LLM report generator. DenseNet-121 achieved a mean test AUROC of 0.7527 on a 28,597-image patient-level CheXpert test set. The central design contribution is the Evidence JSON contract, a structured interface that limits the LLM to claims derivable from classifier outputs and optional Grad-CAM evidence.

Future work includes full 14-label CheXpert evaluation, external validation on MIMIC-CXR and PadChest, reliability-diagram calibration, RAG grounding using approved reporting templates, formal hallucination benchmarking, and a reader study comparing radiologist and patient comprehension of structured AI reports against unstructured baselines. The goal is not to automate radiology but to provide an auditable scaffold around imperfect AI predictions.

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